Name: ____________

Date: ____________

Vectors

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can represent a vector geometrically and in component form.**

RELATED SKILLS

- I — Read and write vector notation and correctly use the terms magnitude, direction, component, and unit vector.
- I — Write a vector in component form.
- A — Add congruence, parallel, perpendicular, and angle markings to a diagram.

☐ **2. I can add, subtract, and multiply vectors by scalars.**

RELATED SKILLS

- I — Perform component-wise vector operations.
- A — Add, subtract, multiply, and divide signed integers.

☐ **3. I can determine vector magnitude, direction, and unit vector.**

RELATED SKILLS

- I — Calculate vector magnitude using the distance formula.
- I — Determine a vector's direction angle with correct quadrant.
- I — Normalize a nonzero vector to obtain a unit vector.
- A — Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

☐ **4. I can resolve a vector into horizontal and vertical components.**

RELATED SKILLS

- D — Write a vector in component form.
- A — Determine a vector's direction angle with correct quadrant.
- A — Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.

☐ **5. I can perform basic operations with vectors in three dimensions.**

RELATED SKILLS

- I — Represent and operate on vectors with three components.
- D — Calculate vector magnitude using the distance formula.
- D — Perform component-wise vector operations.